

HAQ Lab
Hannigan Air Quality Lab



Lesson 2: How Do We Understand Air Quality?

Central AQ Slide Deck

Last Updated September 8, 2026 by Percy Smith

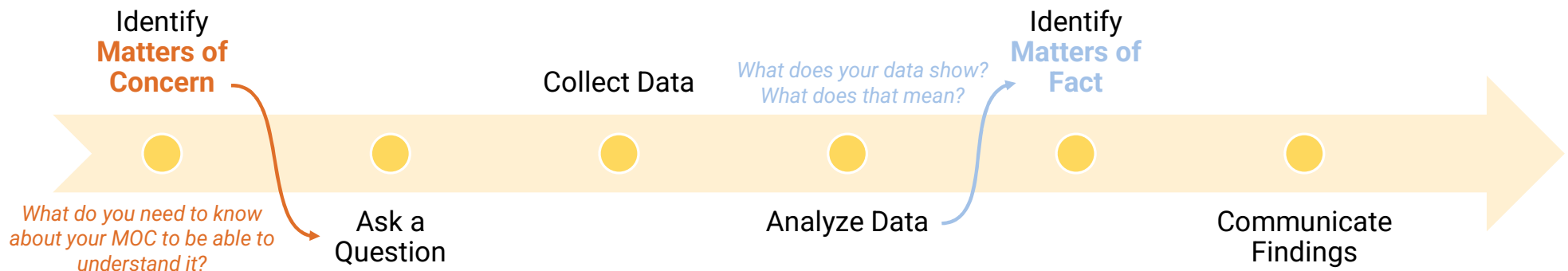


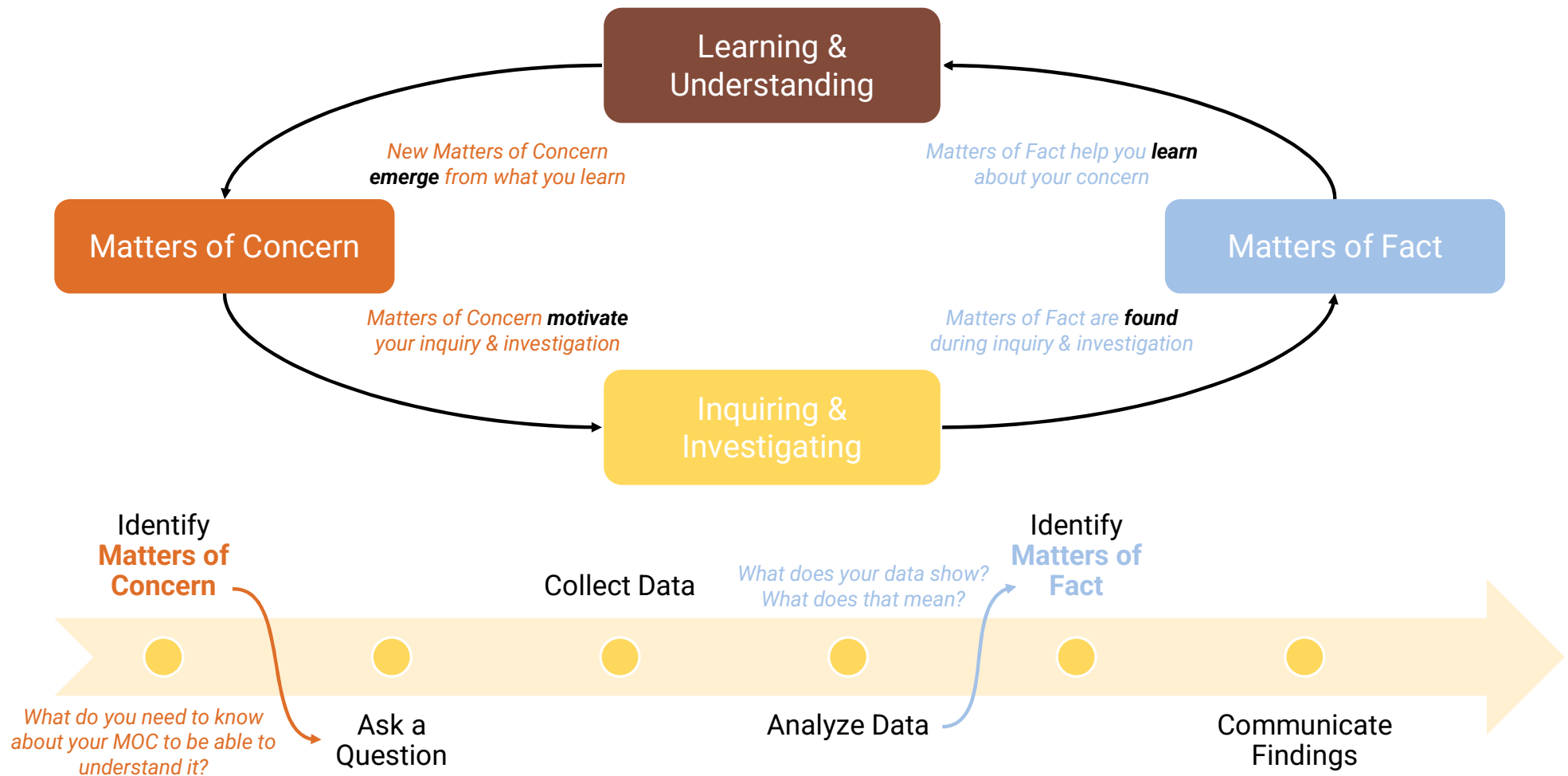
Engineering & Applied Science
UNIVERSITY OF COLORADO BOULDER



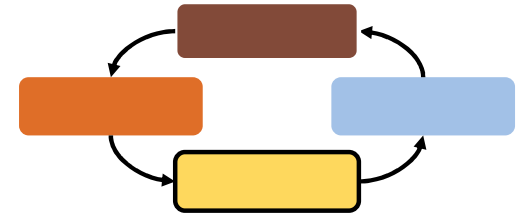
Air Quality InQuiry: MOC & MOF

- **Matters of Concern:** Issues related to a topic that you want to understand
 - Air quality issues that are important to you
- **Matters of Fact:** Conclusions, information & experimental results that define an issue
 - “Knowns” that you use to understand something





Qualitative & Quantitative Data



Quantitative Data

Quantify: to measure or count

Aims to answer “what” & “how much”
with numerical data

Qualitative Data

Qualify: to note qualities or features

Aims to explore “how” & “why” through
descriptive data



Integrating Both!

Quantitative Data

Quantify: to measure or count

Aims to answer “what” & “how much” with numerical data

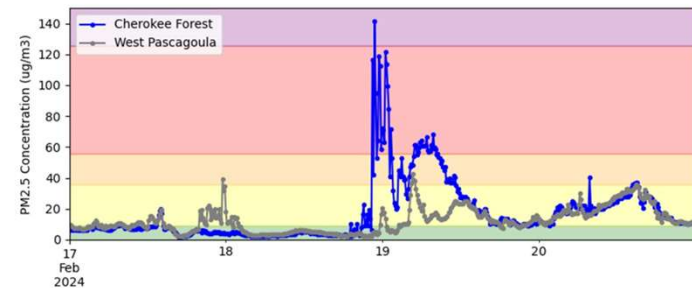


Figure 6. PM2.5 concentrations ($\mu\text{g m}^{-3}$) in the Cherokee Forest subdivision and in West Pascagoula. Elevated PM2.5 concentrations in Cherokee Forest on the evening of February 18th through the 19th correspond to symptom complaints reported by resident participants. Color background on the figure corresponds to the EPA air quality index (AQI) for PM2.5.

Qualitative Data

Qualify: to note qualities or features

Aims to explore “how” & “why” through descriptive data

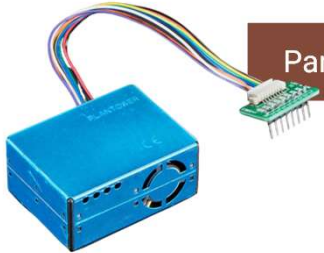
Date and time range of the complaint	Who noticed the smell or felt the symptom?	Where was the reporting person and what were they doing?	What is the smell like (if applicable)?	Smell intensity from 1 (very mild) to 5 (very intense)	What symptoms are felt (if applicable)?	Symptom intensity from 1 (very mild) to 5 (very intense)
2-8-24 7 PM	Myself ☒ Spouse/partner ☐ Child/grandchild ☐ Other:	Truckly garbage outside.	☒ Rotten eggs ☐ Cat urine/ammonia ☐ Garbage ☐ Acid ☐ Plastic burning ☐ Gasoline or fuel ☐ Sweet smell ☐ Other:	5	☐ Headache ☒ Itchy/burning eyes ☐ Skin itching/rash ☐ Dizziness/lightheadedness ☐ Sinus pain ☒ Sore throat ☐ Fatigue ☐ Difficulty breathing ☐ Other:	4
Anything else you would like to share: Strong Smell. Chemical smell.						

Figure 3. An example symptom and odor log entry.





Carbon Monoxide (CO)



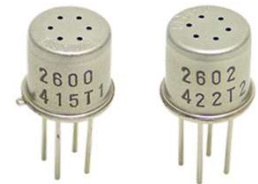
Particulate Matter (PM 2.5)



Carbon Dioxide (CO₂)



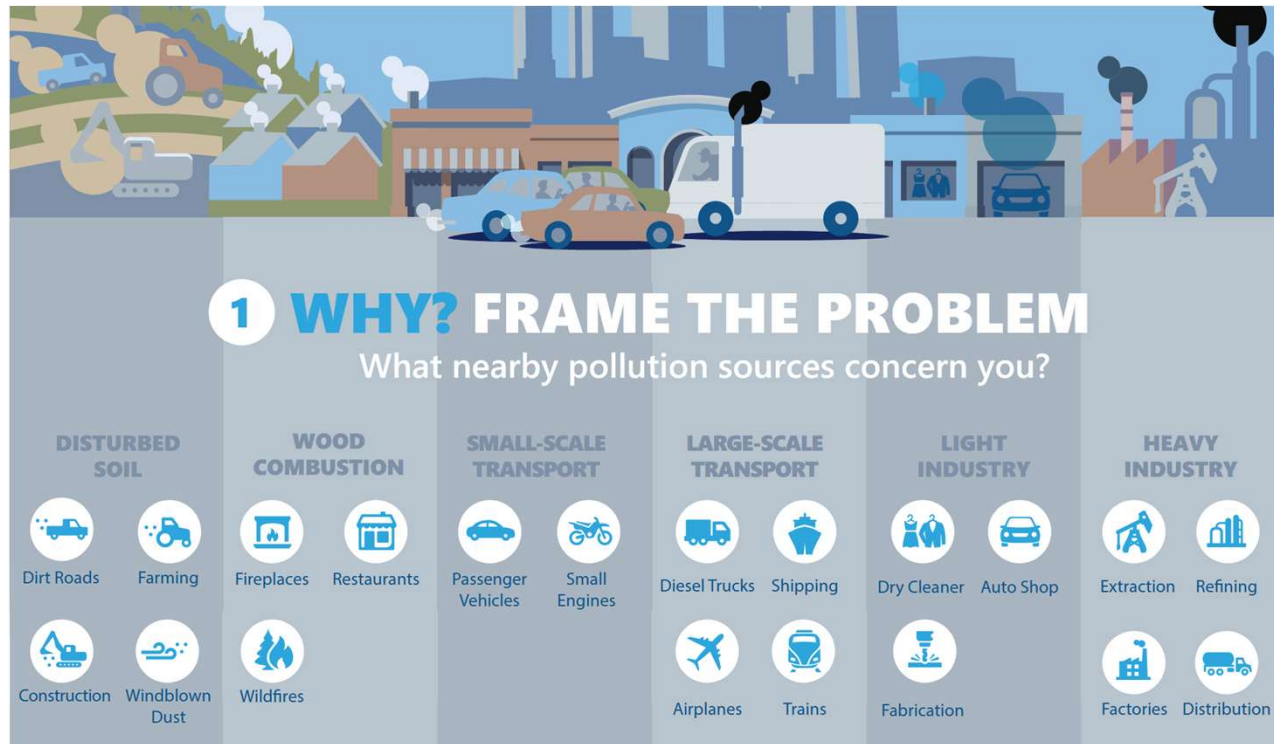
Volatile Organic Compounds (VOC's)



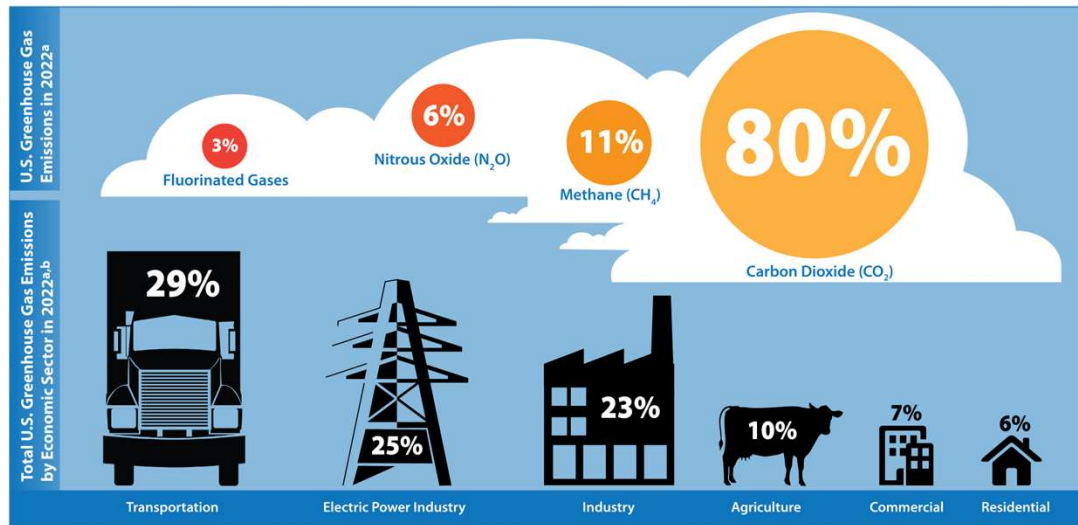
The YPOD



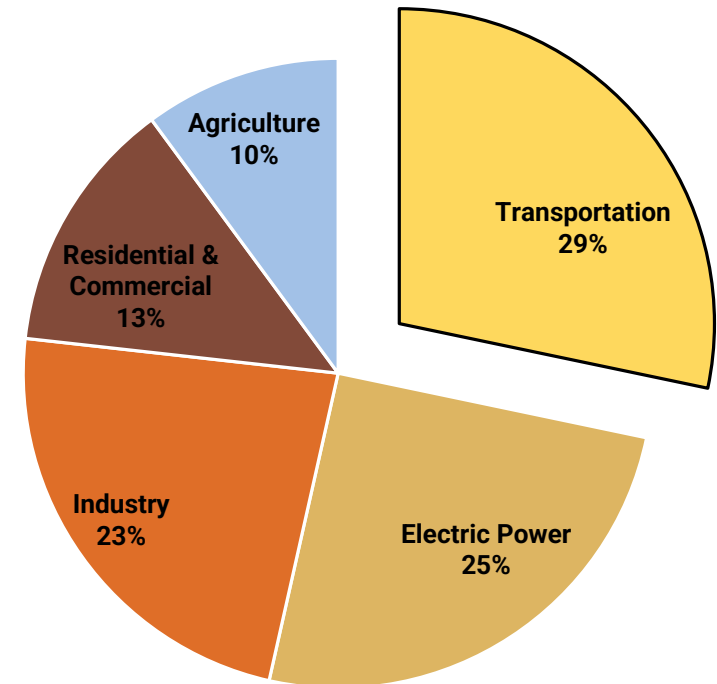
Numerous Air Quality Concerns



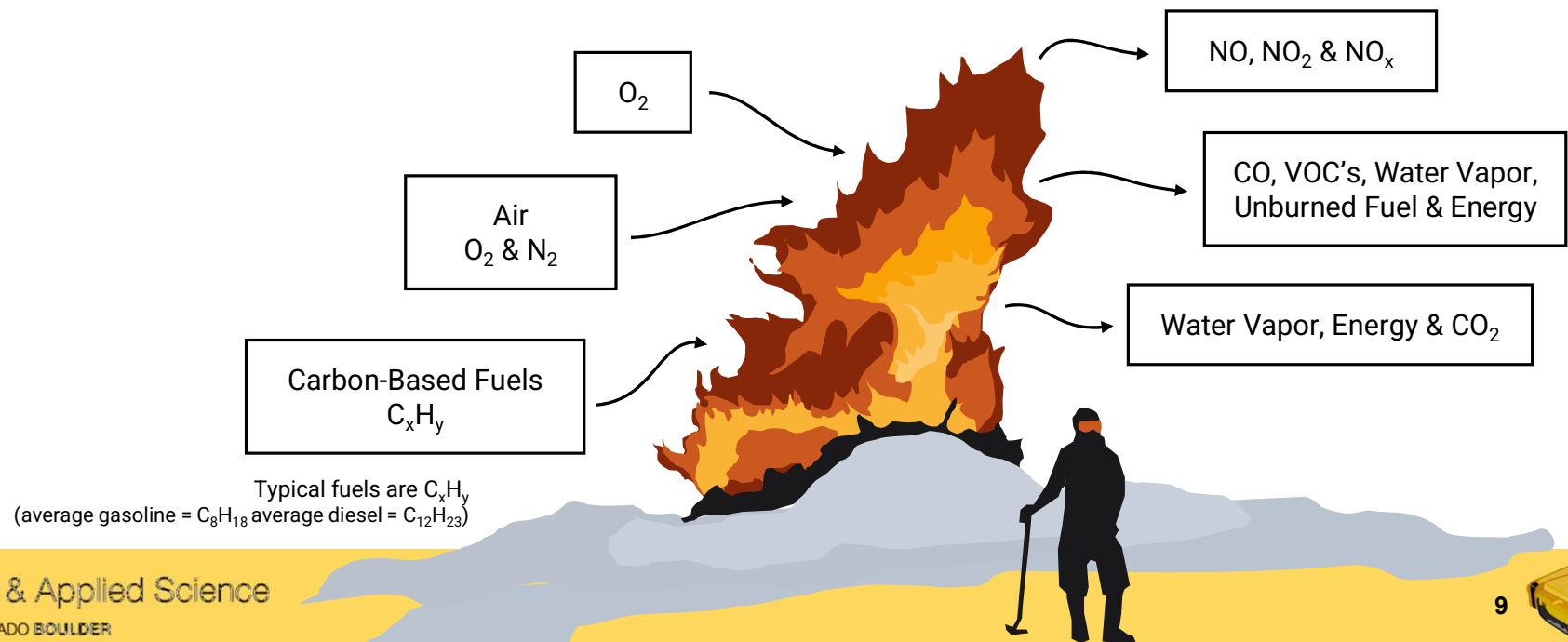
Emissions from Transportation



Total US Greenhouse Gas Emissions by Economic Sector in 2022

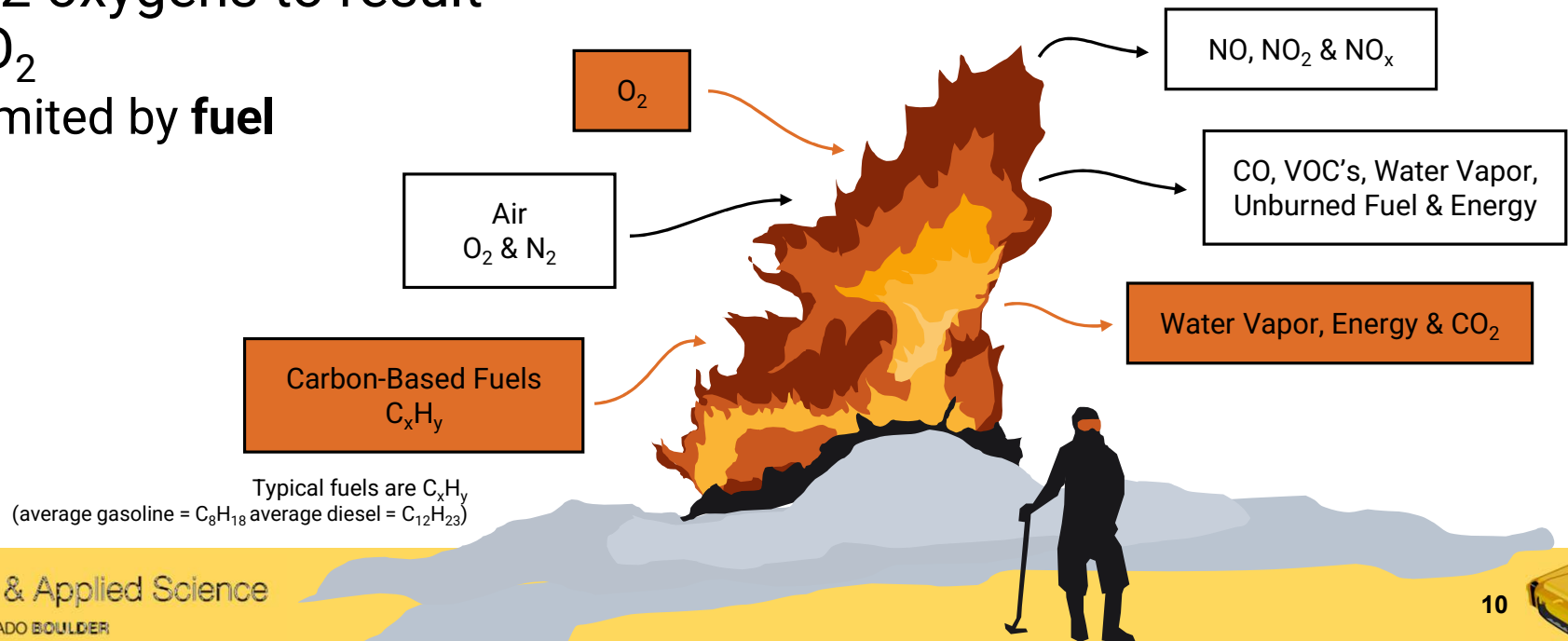


Combustion's 3 Essential Reactions



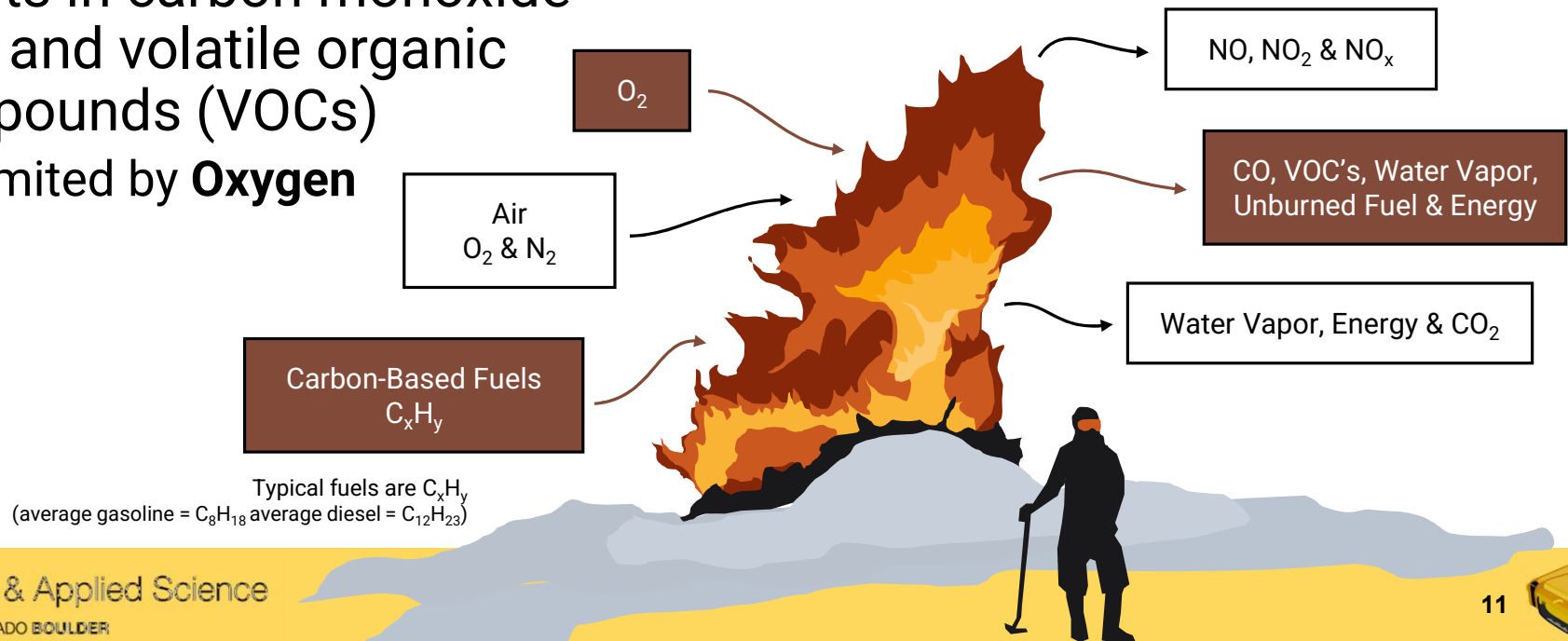
Reaction 1: Complete Combustion

- Requires sufficient oxygen to pair carbon with 2 oxygens to result in CO_2
 - Limited by **fuel**



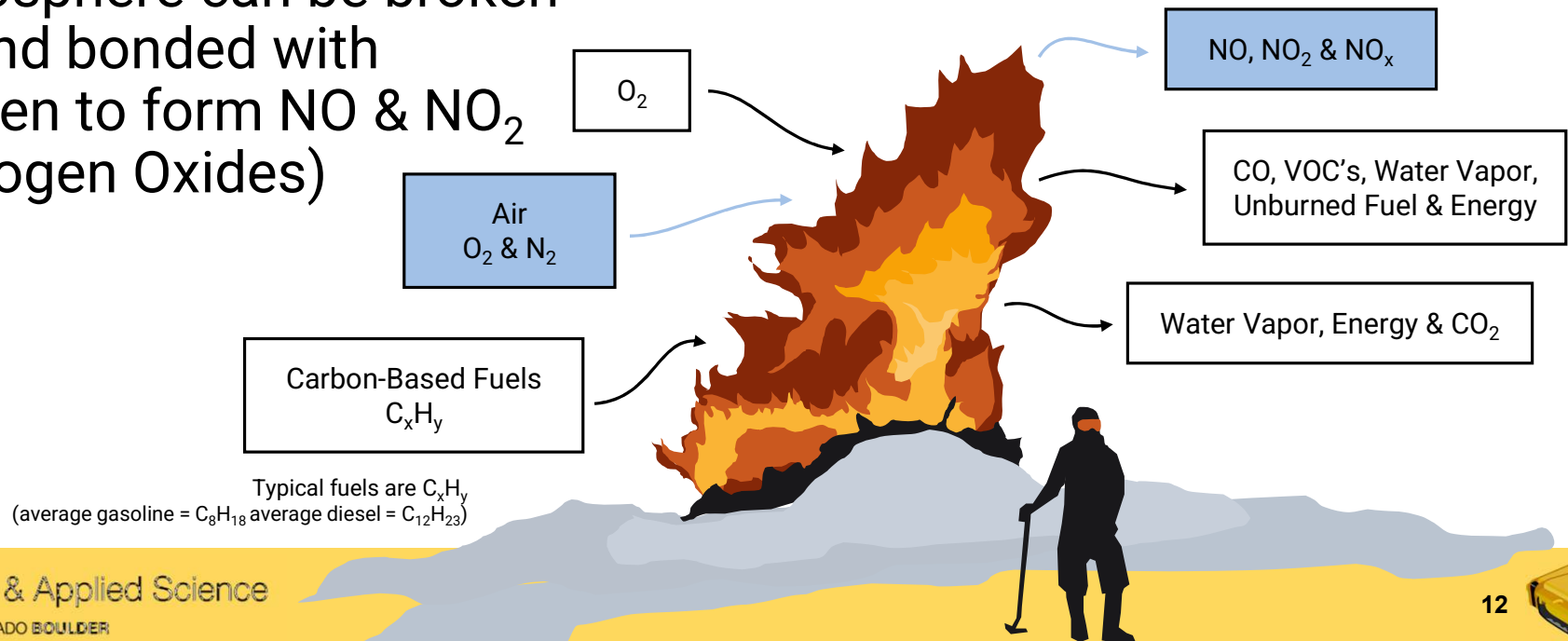
Reaction 2: **Incomplete Combustion**

- Without enough oxygen, combustion is incomplete & results in carbon monoxide (CO) and volatile organic compounds (VOCs)
 - Limited by **Oxygen**



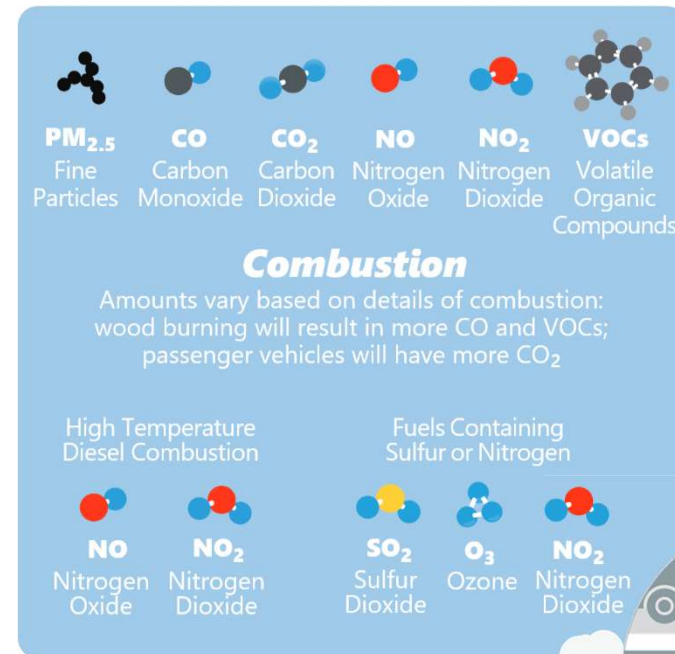
Reaction 3: Thermal NO_x Formation

- During high-temperature combustion, N_2 from the atmosphere can be broken up and bonded with oxygen to form NO & NO_2 (Nitrogen Oxides)



Vehicle Pollutants

- How do vehicles generate particulate matter (PM)?
 - Direct emission from tailpipes from combustion
 - Tire & break wear
 - Dust generation from driving on dirt & asphalt roads
- What is black soot?
 - Dark exhaust emitted from diesel-powered vehicles
 - One of the most harmful emissions for human health
 - Forms from incomplete fuel combustion



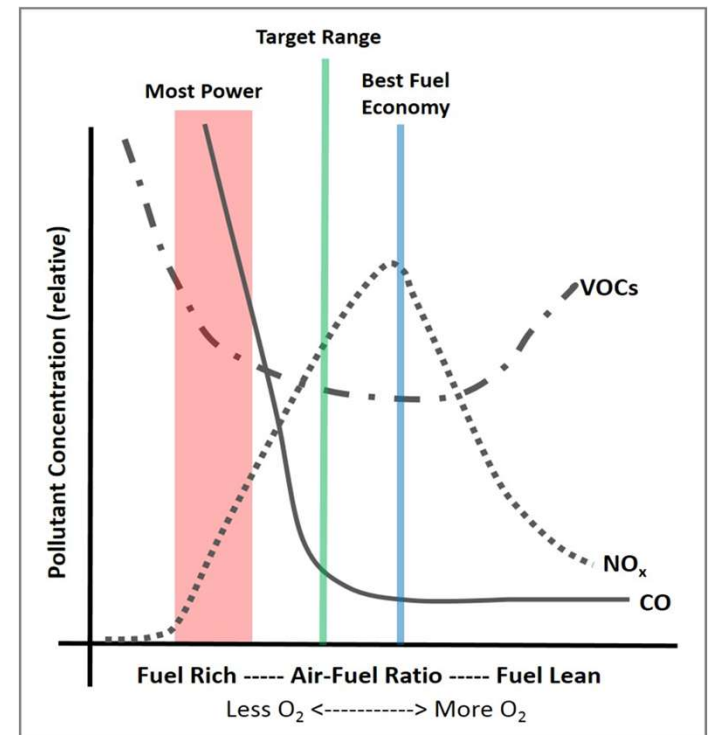
Cars, Buses, Trucks,
Planes, Trains



Vehicular Combustion: Air-Fuel Ratio

Air-Fuel Ratio: Fuel rich v. fuel lean is controlled by how much fuel is sent to the engine

	Fuel Rich	Fuel Lean
What it means	More fuel, less oxygen	Less fuel, more oxygen
Pros	More power available for vehicle	More complete combustion (fewer harmful pollutants)
Cons	Less fuel-efficient and more harmful pollutants (CO, VOCs, unburned fuel)	Less power available, possibly more NO _x produced (if high-temperature combustion)



CO & Air Pollution Control Technology

In 1975, catalytic converters were built to help curb CO emissions from vehicles

Check out this General Motors advertisement from 1975 showing off a new catalytic converter!

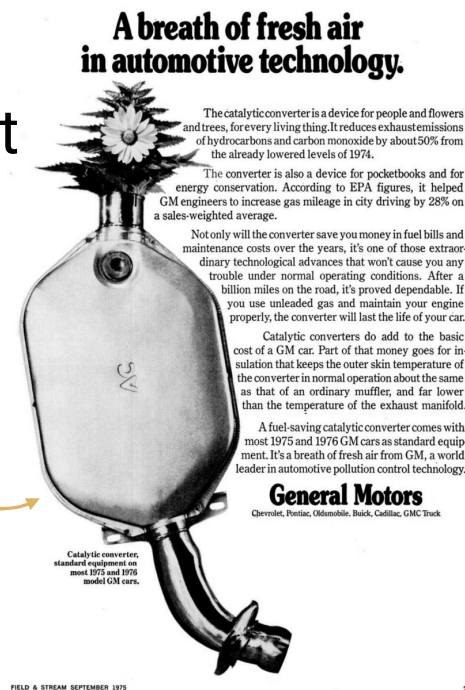
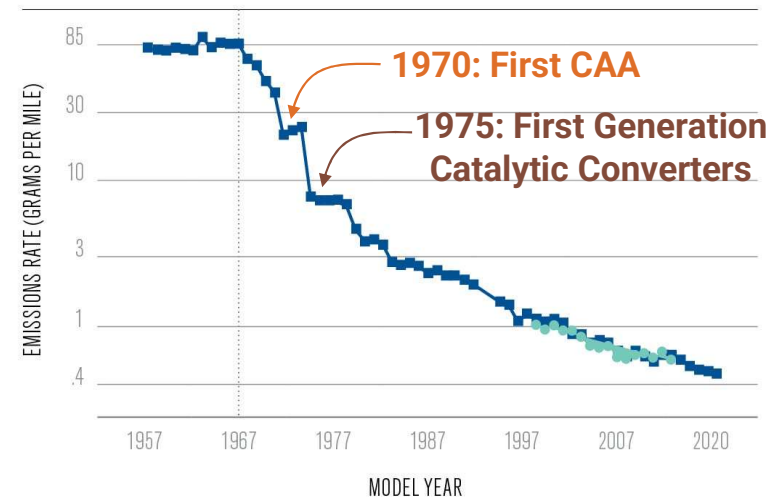
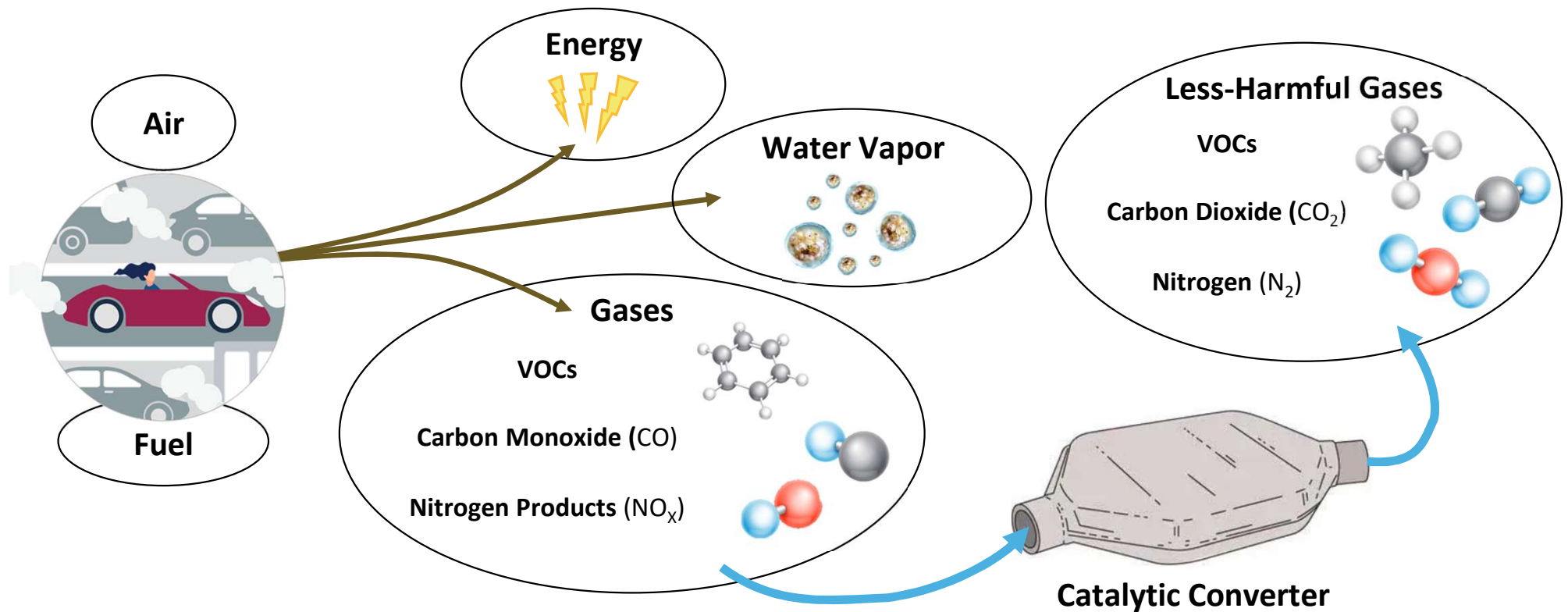


FIGURE 1: MEAN POLLUTION RATES OF NEW U.S. VEHICLES, 1957-2019

(a) Carbon Monoxide (CO)



Catalytic Converters & Pollution



The Tailpipe Experiment

Dependent variables

- CO₂, VOCs, temperature and relative humidity (RH)

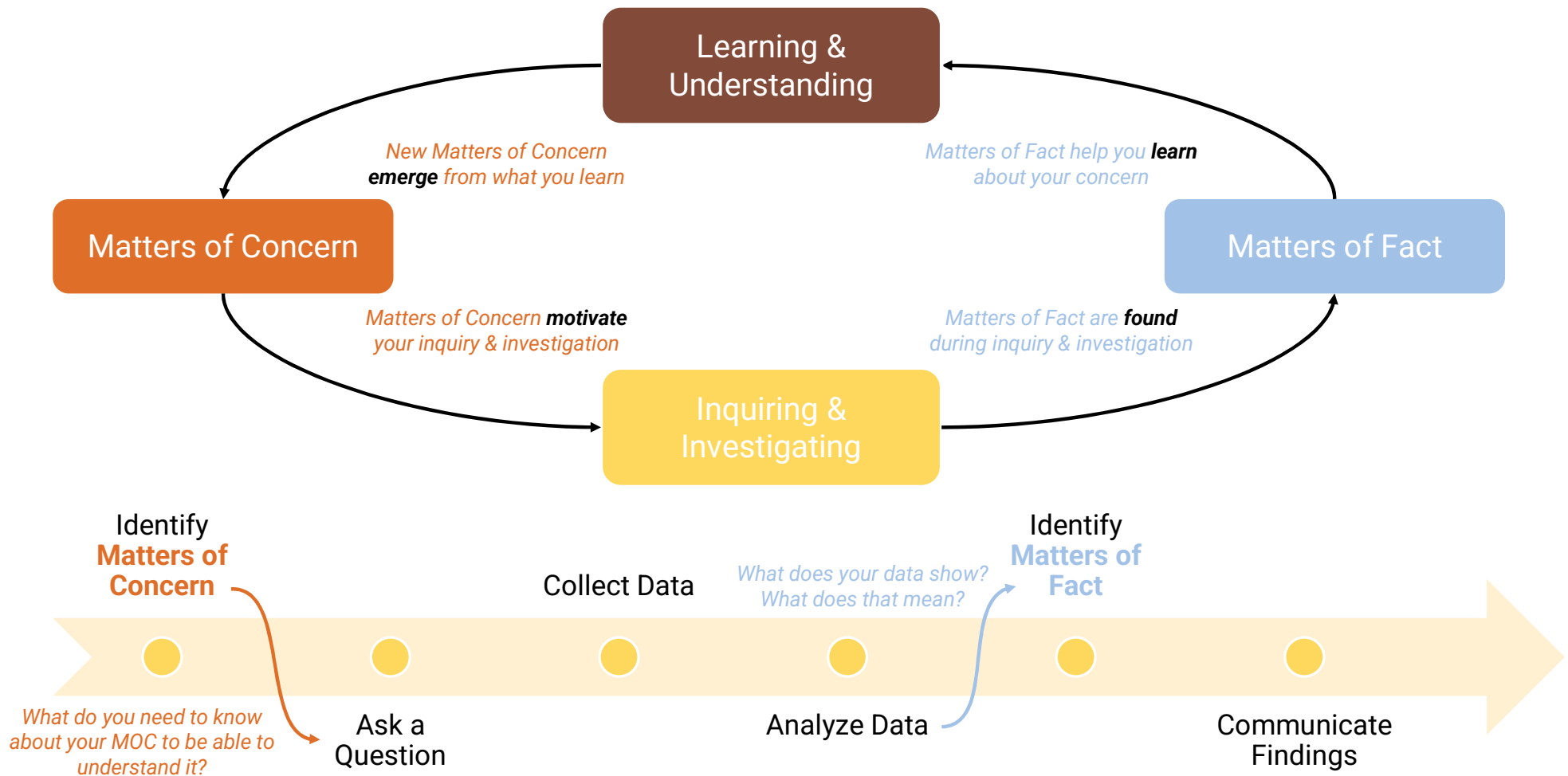
Procedure

- Run monitor outdoors to collect a **baseline measurement**
- Place monitor behind tailpipe, slightly above and straight out
- **Idle vehicle #1**
- Repeat **baseline measurement**
- Replicate setup and **idle vehicle #2**
- Repeat entire series, or finish with baseline measurement

Record on the Activity Data Sheet

- Monitor placement (distance from emission source)
- Start and stop times for emission tests
- Observations (environmental conditions, etc.)
- Vehicle notes (for example, make, model, year, etc.)





If you want to learn more, we recommend these excellent sources!

*Polidori, A., Papapostolou, V., Collier-Oxandale, A., Hafner, H., & Blakey, T. (2021). *Community in Action: A Comprehensive Guidebook on Air Quality Sensors*. South Coast Air Quality Management District. South Coast AQMD's AQ-SPEC. https://www.aqmd.gov/docs/default-source/aq-spec/star-grant/community-in-action-a-comprehensive-guidebook-on-air-quality-sensors.pdf?sfvrsn=10*

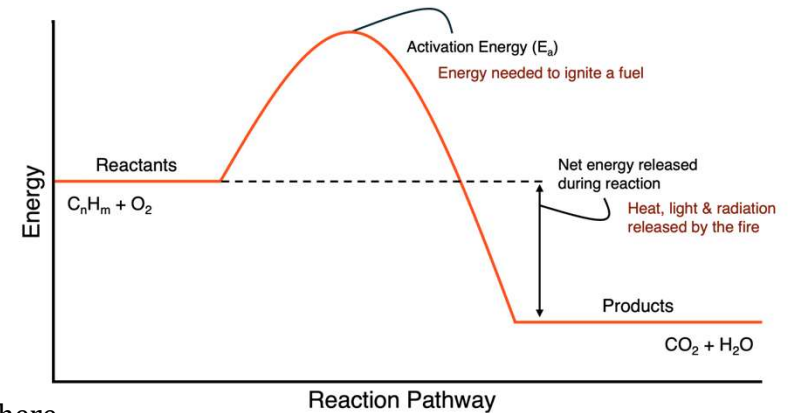
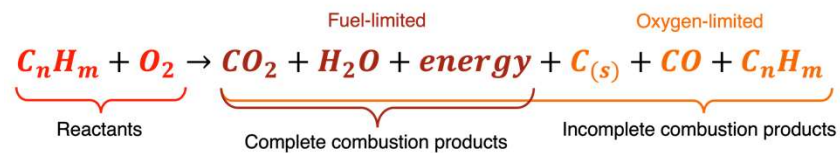
De Greve, T., Malliet, S., Hendriks, N., & Zaman, B. (2022). The Air Quality Lens: Ambiguity as Opportunity to Reactivate Environmental Data. *Proceedings of the 2022 ACM Designing Interactive Systems Conference, DIS '22*, 335–348. <https://doi.org/10.1145/3532106.3533530>

US EPA. (2023). *Greenhouse Gas Inventory Data Explorer*. US EPA. <https://cfpub.epa.gov/ghgdata/inventoryexplorer/#allsectors/allsectors/allgas/select/all>

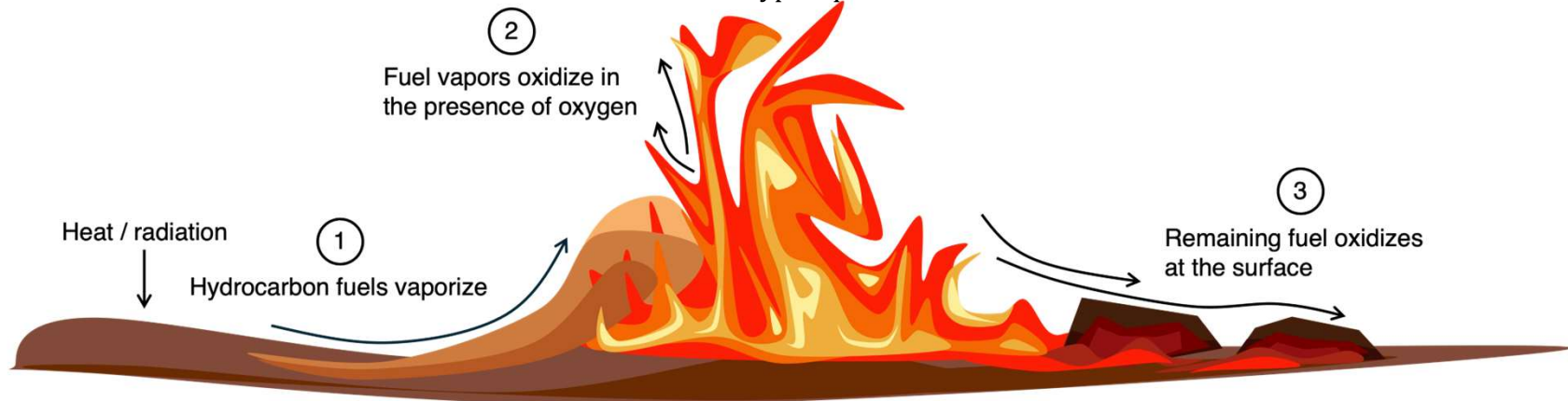
US EPA. (2026, June 10). *Inventory of U.S. Greenhouse Gas Emissions and Sinks* [Reports and Assessments]. US EPA. <https://www.epa.gov/ghgemissions/inventory-us-greenhouse-gas-emissions-and-sinks>



What is Combustion?



Type equation here.



Which vehicle releases more pollution?

Honda CRV **with** Catalytic Converter



Motorcycle with **no** Catalytic Converter



As an example of what catalytic converters can do, consider this experiment that the HAQ Lab ran



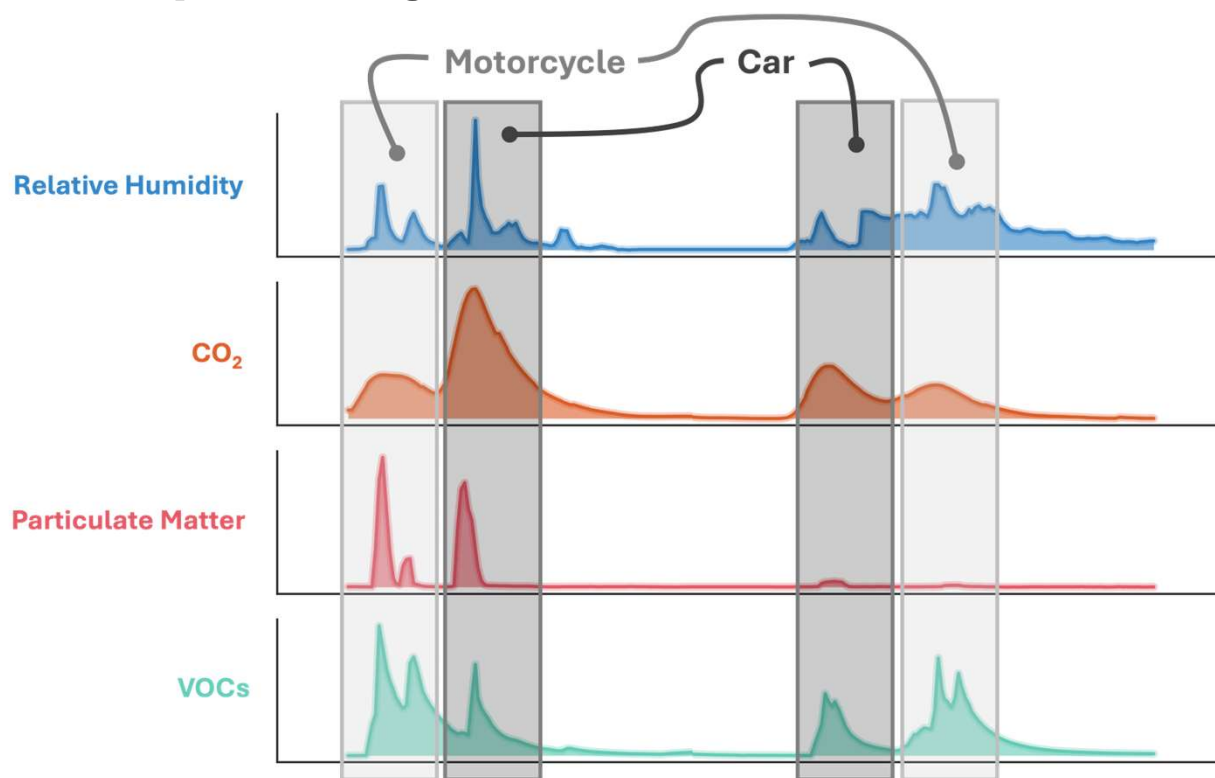
Slide 21

EC1

(or bonus question: which one emits more pollution per passenger mile or per liter of fuel burned?) Normalizing the emissions is important when comparing different "services"

Evan Coffey, 2026-09-08T22:52:39.758

Comparing the Vehicles Directly...



The Honda
emits less
toxic pollution!

Slide 22

EC1

CO would be nice on here

Evan Coffey, 2026-09-08T22:53:11.539